

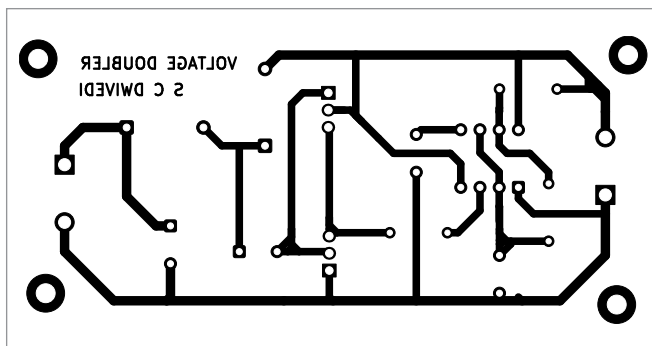


Fig. 3: Actual-size PCB

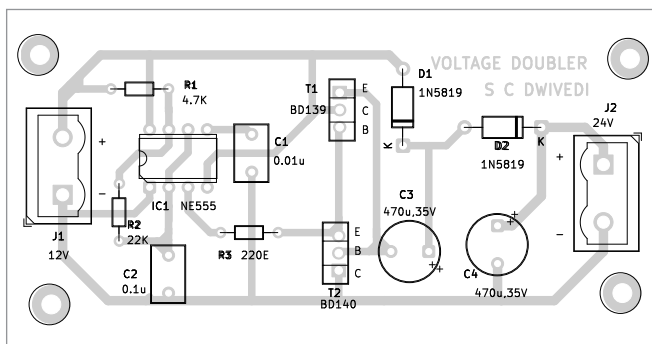


Fig. 4: Components side layout

suitable enclosure for reliable operation and protection.

Alternatively, the system may be assembled on a general-purpose PCB or a small perforated board. Begin the assembly by mounting the IC socket for the NE555 timer, followed by resistors R1 and R2, and capacitors C1 and C2 associated with the oscillator section. Ensure proper soldering and correct component placement for stable oscillator operation.

Next, mount transistors BD139

(T1) and BD140 (T2), ensuring the correct orientation of their collector, base, and emitter terminals. For applications requiring higher output current, small heat-sinks may be fixed to the transistors to improve heat dissipation and prevent overheating.

After completing the driver stage, solder Schottky diodes D1 and D2, taking care to observe their polarity. Then solder electrolytic capacitors C3 and C4 with the correct polarity. Finally, connect input connector J1 for the 12V DC supply and output connector J2 for the boosted DC output.

Once assembly is complete, thoroughly inspect the PCB for solder bridges, dry joints, incorrect wiring, or reversed component polarity before applying power.

For testing, connect a regulated 12V DC supply to connector J1. Using a digital multimeter, measure the voltage across output connector

PARTS LIST

Semiconductors:

- IC1 - NE555 timer IC
- T1 - BD139 NPN transistor
- T2 - BD140 NPN transistor
- D1, D2 - 1N5819 Schottky diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 4.7k Ω
- R2 - 22k Ω
- R3 - 220 Ω

Capacitors:

- C1 - 0.01 μ F ceramic capacitor
- C2 - 0.1 μ F ceramic capacitor
- C3, C4 - 470 μ F, 35V electrolytic capacitor

Miscellaneous:

- J1, J2 - 2-pin terminal connector
- General-purpose PCB
- Suitable enclosure
- IC base for NE555 (optional)

J2. Under no-load or light-load conditions, the system should produce an output voltage of approximately 20V to 24V DC.

EFY note. This circuit also operates with lower input voltages. For example, a 6V input can produce approximately 12V, a 9V input can generate around 17V, and a 12V input can deliver nearly 24V. The actual output voltage may vary depending on the load current, input supply conditions, and component tolerances. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-6v-9v-and-12v-universal-voltage-doubler>

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